

Experimental Auctions

Learning and Bounded Rationality

Daniel Stephenson

Texas A&M University

A Simple Auction Experiment

- We are going to conduct a simple auction experiment.

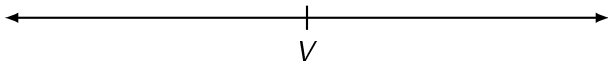
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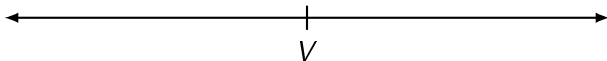
- We are going to conduct a simple auction experiment.
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- We are going to auction off a prize with a value of V cents.
- The value V is somewhere between 100 and 300.

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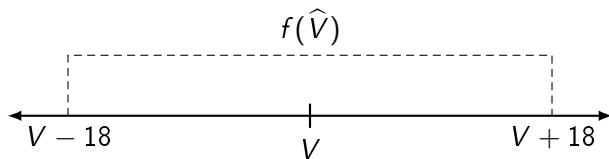
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- Your estimate $\hat{V}$ is uniformly distributed on $[V - 18, V + 18]$.

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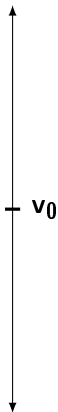
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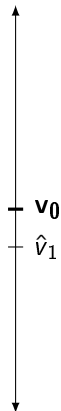
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- Try to maximize your profits.

Sealed Bid Common Value Auctions



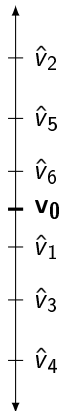
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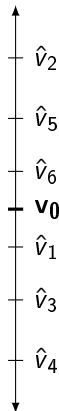
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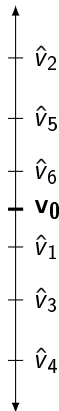
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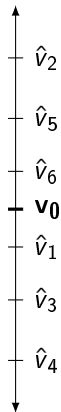
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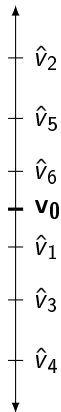
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 - i.e. $E\{\hat{v}_i\} = v_0$

The Winner's Curse



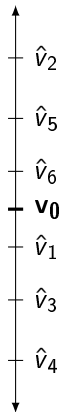
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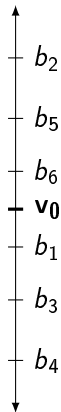
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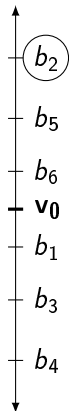
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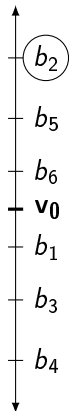
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- Large bids will correspond to large estimates.
- The highest bidder is likely to overbid.
- Rational bidders should shade their bids:
 - i.e. consider the expected value of v_0 *conditional on winning the auction*.

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- This would help explain why subjects tend to overbid.

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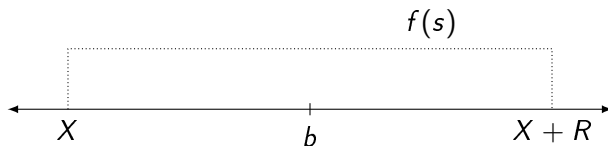
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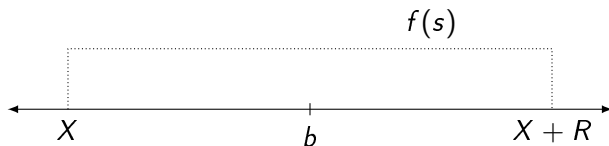
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- If overbidding still occurs in the no curse treatment, then subjects may get utility from winning.

Bilateral Bargaining



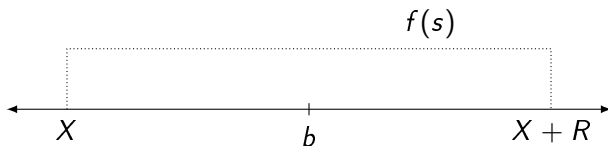
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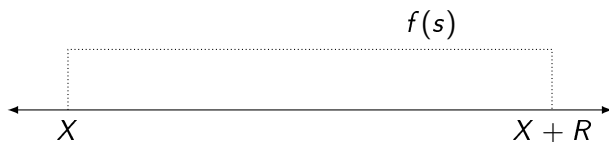
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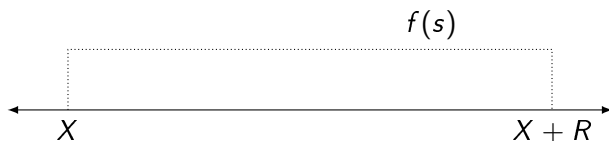
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- The buyer issues a bid b . The seller will only accept the bid when $s < b$.

Naive Bidding



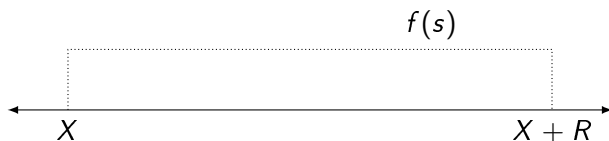
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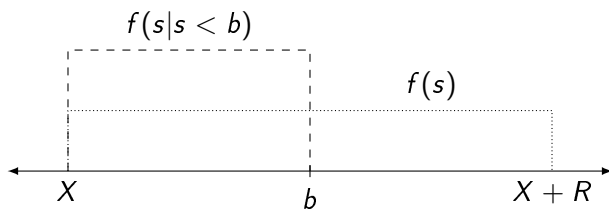
- $P(s < b) \cdot [E(v) - b]$ the naive objective
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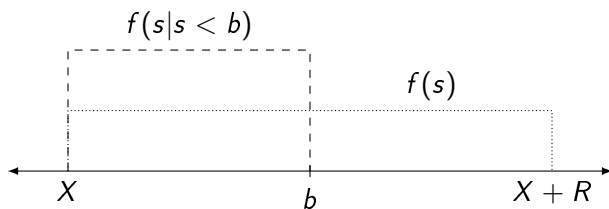
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Rational Bidding



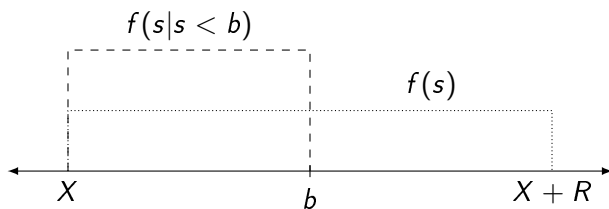
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- $b_R = \left(\frac{1}{2 - M} \right) X$ the rational bid

Alternative Treatments and Predictions

Treatment	Parameters			Bids		Expected Earnings	
	R	X	M	b_R	b_N	for b_R	for b_N
Winner's Curse	4.5	1.5	1.5	3	3.56	0.25	0.21
No Curse	2.0	1.0	1.5	2	2.00	0.25	0.25
Loser's Curse	0.5	0.5	1.5	1	0.81	0.25	0.21

Average Observed Bidding

	Average Bid		
	Winner's Curse	No Curse	Loser's Curse
Rational	3.00	2.00	1.00
Naive	3.56	2.00	0.81
Data Aggregate	3.78	2.03	0.74
$n = 50$	(0.58)	(0.21)	(0.08)

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- Holt and Sherman reject the “utility of winning” hypothesis.
- The naive bidder model is consistent with the data.

Experienced Bidders

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- Experienced subjects had participated in four prior sessions.
- To cover potential losses, bidders had a starting capital of \$10.

Market Earnings for Experienced Bidders

n	Observed	Theoretical	Difference	# auctions
4	2.28 (0.52)	6.95 (0.49)	3.67 (0.35)	149
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- In contrast, inexperienced bidders typically had negative earnings.
- Kagel and Richard conclude that experience can significantly mitigate the winner's curse.

Individual Ability and Market Selection

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- Bankrupt subjects may be less interested in future experiments.
- Similar “market selection” dynamics may take place in auctions outside the laboratory.

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- This is lessened by randomly increasing or decreasing the initial endowment.

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- Previous estimates may have confounded learning with market selection.

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- Ascending Bid: bidders successively place publicly visible bids.

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- In experimental auctions, seller revenue is typically higher under the sealed bid mechanism.
- However, the vast majority of auctions outside the laboratory continue to use the ascending bid mechanism.

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- In each of these rounds, bidders choose which auction to participate in.

Observed Participation and Revenue

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- Endogenous entry may help explain why most auctions continue to use the ascending bid mechanism.

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 - Less than $\frac{1}{4}$ of Amazon auctions have bids in the last hour.
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- Ariely, Ockenfels, and Roth (2005), hypothesize that Amazon's “soft closing” rules will mitigate these incentives.

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- It can be difficult to draw solid conclusions about the effects of eBay's "hard closing" rules from field data.
- Ariely, Ockenfels, and Roth (2005) conduct experiments to examine bidding behavior under alternate closing rules.

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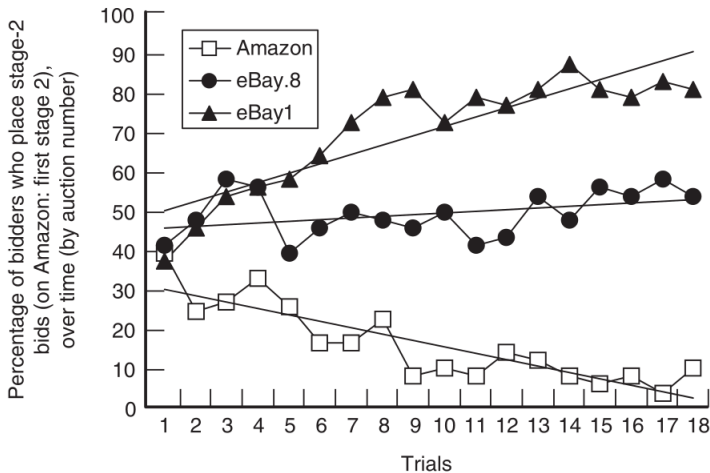
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- Each treatment has two phases.
 - Phase 1: A sequence of ascending price bidding periods. This phase is the same in every treatment.
 - Phase 2: Closing period. Rules depend on the treatment employed.

Observed Last Minute Bidding Over Time



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- Probabilistic last minute transmission reduces the effects of hard closing.
- The laboratory data is consistent with the bidding behavior observed in the field.
- The combination of laboratory data and field data provides stronger evidence for the effect of hard closing rules than either could provide individually.

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